

B.Sc. 6th Semester (Honours) Examination, 2025

COMPUTER SCIENCE

Course ID: 61511

Course Code: SH/CSC/601/C-13

Artificial Intelligence (Theory)

(CBCS NEW)

Time: 1 Hr 15 Min

Full Marks: 25

*The figures in the right-hand margin indicate marks.
Candidates are required to give their answer in their own words as far as practicable.*

Answer the questions unit wise as instructed

UNIT- I

1. Answer any five of the following questions:

5 × 1 = 5

- a) What do you mean By Artificial intelligence?
- b) Construct a truth table for the expression $((p \wedge q) \wedge (p \vee q))$.
- c) Write down the domain of AI.
- d) What do you mean by chronological Backtracking in AI?
- e) What is Heuristic Search?
- f) What do you mean Control strategy?
- g) Define Agent.
- h) What is Predicate Logic in AI?

2. Answer any two of the following questions:

2 × 5 = 10

- a) How many possibilities are there in tic-tac-toe? Write down the rules of Control Strategy. 1+4=5
- b) Write down the difference between propositional Logic and predicate Logic.
- c) Write down the steepest hill climbing algorithm.
- d) What are the four ways of knowledge representation? Explain any one.

2+3=5
[Turn Over]

3. Answer any two of the following questions:

1 × 10 = 10

a) Write down the AO* algorithm in AI? Difference between A* and AO* algorithm. write the application of AO* algorithm.

[5+3+2=10]

b) How AI and game theory are related? Write, down the properties of min-max algorithm? Compare between min-max and alpha-beta pruning algorithm?

[5+3+2=10]

B.Sc. 6th Semester (Honours) Examination, 2025

COMPUTER SCIENCE

Course ID: 61511

Course Code: SH/CSC/601/C-13

Artificial Intelligence (Theory)

(CBCS OLD 2017-18)

Time: 1 Hr 15 Min

Full Marks: 25

*The figures in the right-hand margin indicate marks.
Candidates are required to give their answer in their own words as far as practicable.*

Answer the questions unit wise as instructed

UNIT- I

1. Answer any five of the following questions:

5 × 1 = 5

- a) What do you mean By Artificial intelligence?
- b) Write the application of Artificial Intelligence?
- c) What is semantics Net?
- d) What do you mean by Backtracking in AI?
- e) What is universal and existential Quantifiers?
- f) What do you mean Constraint Satisfaction Problem (CSP)?
- g) Define Turing test.
- h) What is Predicate Logic in AI?

2. Answer any two of the following questions:

2 × 5 = 10

- a) Write down the rules of control strategy. what do you mean by Production system in AI. 4+1=5
- b) What are the different type of knowledge representation in AI. Explain any one. 2+3=5
- c) Distinguish simple hill climbing from steepest hill climbing algorithm?
- d) Write down the BFS and DFS Algorithm? 2+3=5

[Turn Over]

3. Answer any two of the following questions:

1 × 10 = 10

- a) Write down the algorithm to build tic-tac-toe in AI? How many possibility are there in the tic-tac-toe? what do you mean by game theory? 5+3+2=10
- b) Explain the water jug problem in AI? Write down the important requirement for control Strategies. Name the rules pf Production System in AI. 5+3+2=10

BANKURA SAMMILANI COLLEGE
INTERNAL ASSESMENT EXAMINATION-2024-25
Computer Science (H)

Course ID-61511

Course Title: Artificial Intelligence

Time: 40 Min

Full Marks: 10

1. Answer any five of the following question:

- a) What is agent in AI? Type of agent.
- b) Difference between BFS and DFS.
- c) Write down the limitation of hill climbing.
- d) What are the rules of control strategies?
- e) What is Heuristic Search?
- f) Write down the difference between informed and uninformed search?
- g) Write the goals of AI?
- h) What do you mean by Turing Test?

B.Sc. 6th Semester (Honours) Examination, 2025

COMPUTER SCIENCE

Course ID: 61521

Course Code: SH/CSC/601/C-13

Artificial Intelligence (Practical)

(CBCS OLD 2017-18)

Time: 2 Hours

Full Marks: 15

The figures in the right-hand margin indicate marks.

Candidates are required to give their answer in their own words as far as practicable.

(LNB + Viva = 05 Marks, Experiment = 10 Marks)

Answer any one of the following questions:

1 × 10 = 10

1. Write a prolog program to implement CD of two numbers.
2. Write a prolog program to find the minimum of two numbers.
3. Write a prolog program to calculate, the nth Fibonacci numbers.
4. Write a prolog program to calculate the sum of two numbers.
5. Write a prolog program to implement max list (List, Max) so that Max is the greatest of two numbers in the list of number list.
6. Write a prolog program to append two lists.
7. Write a prolog program to implement palindrome (list).
8. Write a prolog program to implement max (X, Y, Max) so that Max is the greatest of two numbers X and Y.

[Turn Over]

BANKURA SAMMILANI COLLEGE
BSC (HONS) 6TH SEMESTER MOCK-TEST-2025

Subject: Computer Science

Course ID: 61511

Course Title: Artificial Intelligence

Full Marks: 25

Time-1Hr 15 mins

1. Answer any five question from the following: 5x1=5

- a) What is constraint Satisfaction Problem (CSP)?
- b) What is meant by semantics Net?
- c) Construct a truth table for the expression $(p \wedge (\sim p \vee q))$
- d) What is Turing testing in AI?
- e) Translate the sentence - "Mangoes are sweet" into a formula in predicate logic.
- f) What do we meant by universal and existential Quantifier?
- g) What is meant by knowledge inference?
- h) Write one difference between a frame and a script.

2. Answer any two of the following questions: (5x2=10)

- a) Explain BFS and DFS in AI with examples.
- b) Distinguish between informed and uninformed search technique in AI.
- c) What are the limitations of hill climbing algorithm? Write down the difference between simple Hill climbing and Steepest Hill climbing? 3+2
- d) What is FOPL? Write the following expressions in FOPL: 2+3
 - i. Everyone loves Mary
 - ii. Everyone who sees Mary loves Mary
 - iii. Every Person who loves Mary is happy.

3. Answer any one of the following questions: (1x10=10)

- a) Compare between the min-max and alpha-beta pruning algorithm. Write down the A* algorithm in AI. 5+5
- b) Write down the algorithm to build tic-tac-toe in AI. Discuss about various techniques of knowledge representation in AI? 5+5